



SAP Digital Supply Chain

SAP IBP2311 – What's New Demand Planning

Agenda

Demand

New grouping of algos in UI
Curve Based Forecast
Enhancements in Outlier correction

Demand Sensing

Multiple Launch Dimensions

Product Lifecycle Management

New Guided Procedure
Reminder - Deprecation of external Master Data

General

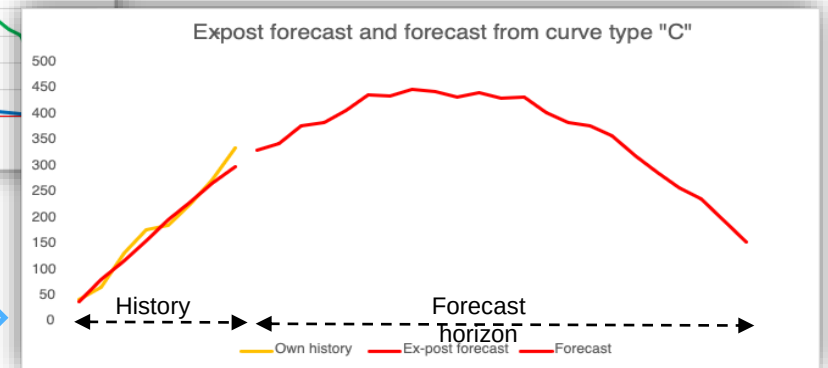
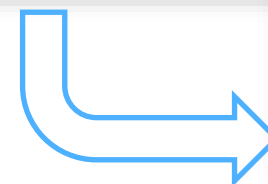
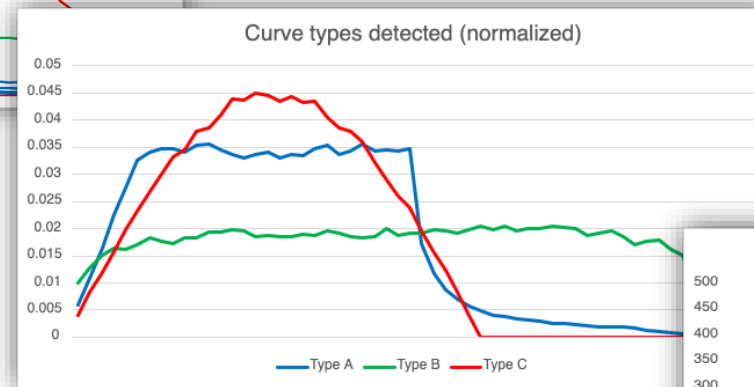
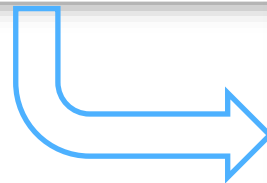
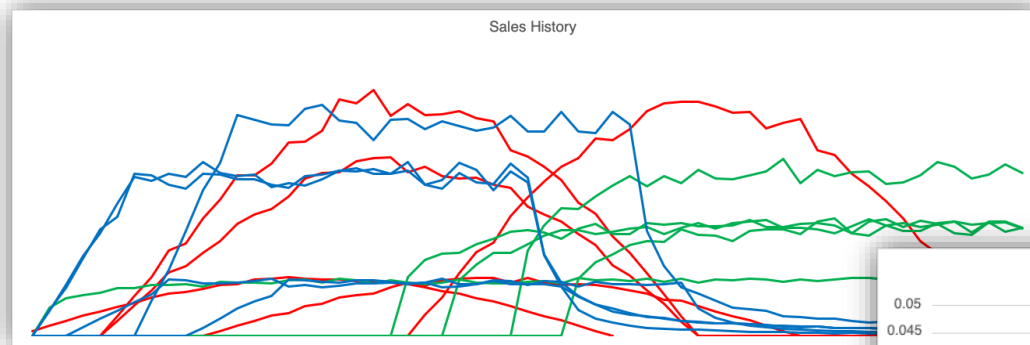
Improved generation of missing planning objects

A blue geometric shape, resembling a parallelogram or a trapezoid, is positioned on the left side of the slide. It has a diagonal line running from the top-left corner to the bottom-right corner.

Demand Curve Based Forecast

What is curve-based forecasting?

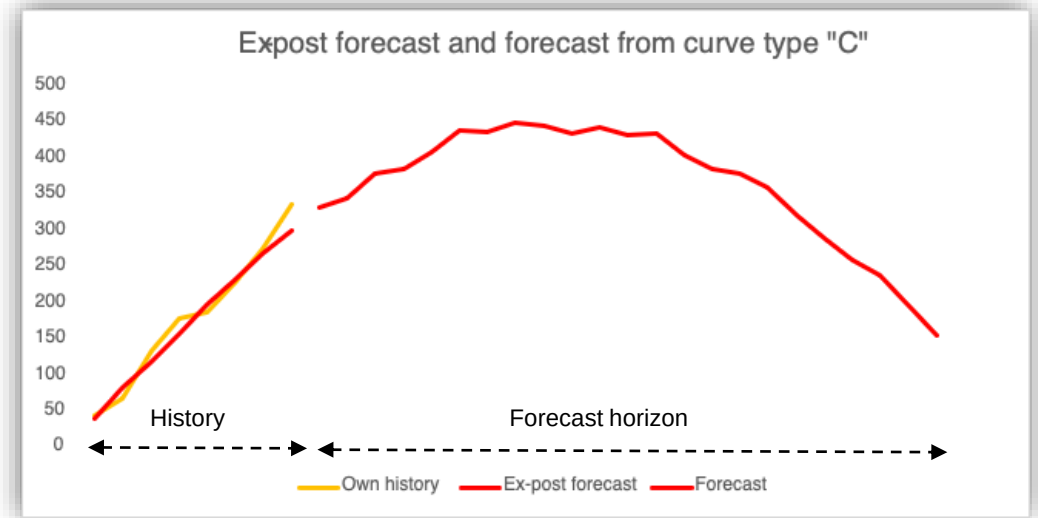
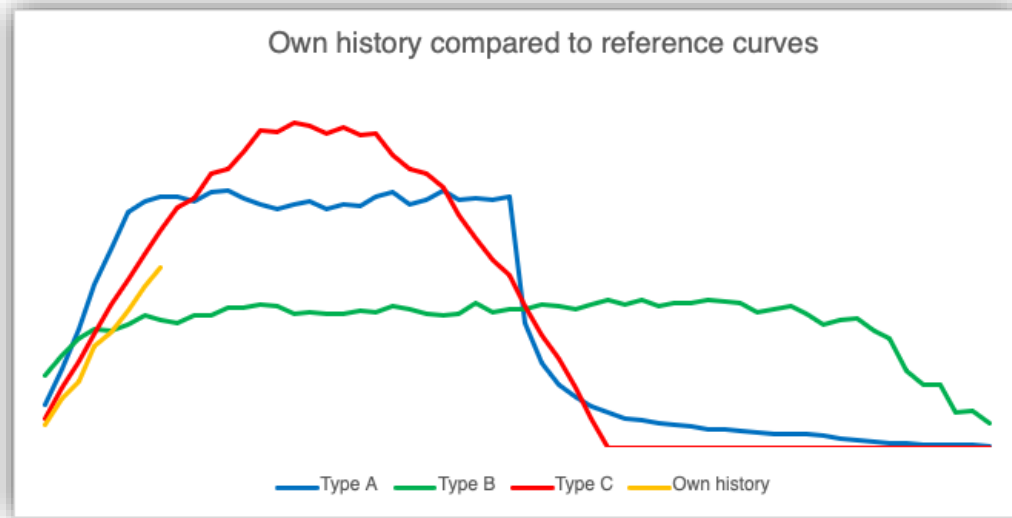
A new approach to predict sales
based on frequent life cycle patterns
observed in the database



Curve-based forecasting: clustering phase



Curve-based forecasting: prediction phase



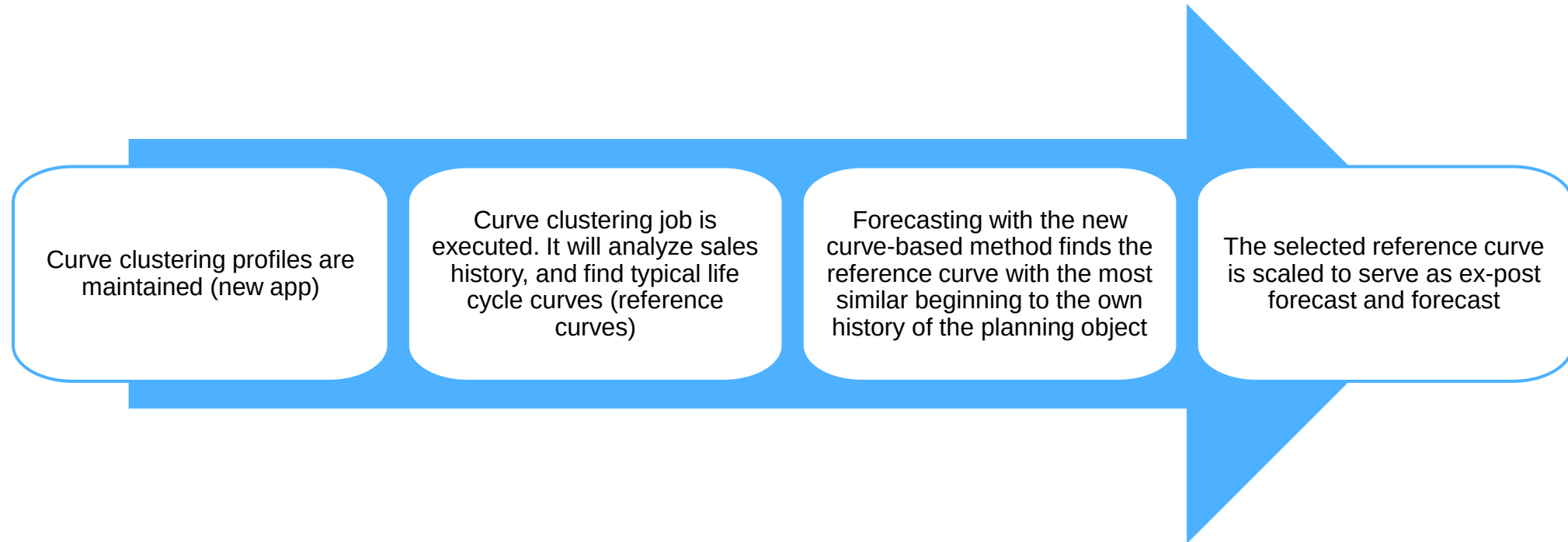
Reference curves long enough to cover [own history + the forecast horizon] are looked up

The selected reference curve is upscaled to the level of own history

The most similarly beginning reference curve is selected (curve type "C" above)

The historical part of the upscaled curve is used as ex-post forecast, the rest is the forecast

Curve-based forecasting: How to use it



Demo

Curve-based forecasting: Manage curve clustering profiles

Input key figure, periodicity and aggregation level must **match the settings of the forecast run**

Specify **number of clusters**, or a range for the system to find the optimal value in

Execute the profile with **job template "Curve Clustering"** before the forecast run

Curve Clustering

1 Template Selection 2 Scheduling Options

3. Parameters

Parameter Section

General

Planning Area: Adam's Demand Forecast PA (ADAM) ×

Profile: ServicePartsMonthly8-20 ×

Planning Filter: CBF forecast (1667353666) ×

Current Period

< SAP Curve Based Forecast Profile ▾ Search In: "Apps" 🔍

ServicePartsMonthly8-20

ADAM

General Settings Parameters Administrative Information

Profile Name: ServicePartsMonthly8-20

Description: Service Parts 8-20 clusters

Planning Area: ADAM

Parameters

Input Key Figure: * ACTUALSQTY ⓘ

Calculation Level: PRDID × LOCID × ⓘ

Calculate Number of Clusters: ☒

Maximum Number of Clusters: 20

Periodicity: * Month ⓘ

Unit of Measure: * EA ▾

Number of Clusters: -----

Minimum Number of Clusters: 8

Administrative Information

Created By: _SAP1530800

Created On: 09/15/2023, 17:08:15

Changed By: -

Changed On: -

Curve-based forecasting: Curve-based method

Specify a **long history horizon**: it should cover the history of as much of the planning objects as possible

Enable **Disregard Leading Nulls** in order to correctly find the first period of sales

If you executed multiple clustering profiles on the level of the forecast run, the system will pick the one based on the **Profile Selection Method** setting

The screenshot displays the SAP 'Manage Forecast Model' interface for 'ServicePartsMonthly'. The 'FORECASTING STEPS' tab is active, showing 'Overall Parameters' and 'Algorithms'. Under 'Algorithms', the 'Curve-Based Method' is selected and highlighted with a red box. A dropdown menu for 'Profile Selection Method' is open, showing options: 'Last Run' (selected), 'Most Extensive', and 'Most Distinct Clusters'. Other visible settings include 'Main Input for Forecasting Steps' set to 'Actuals Qty', 'Target Key Figure for Forecast' set to 'Statistical Fcst Qty', and 'Utilize Multiple Forecasts' set to 'Choose Best Forecast'.

< **SAP** Manage Forecast Model ▾

ServicePartsMonthly

Planning Area ADAM (Adam's Demand Forecast PA)

GENERAL PREPROCESSING STEPS **FORECASTING STEPS** POSTPROCESSING STEPS

Overall Parameters

Main Input for Forecasting Steps: * Actuals Qty

Target Key Figure for Forecast: * Statistical Fcst Qty

Algorithms

> External Algorithm

▾ **Curve-Based Method**

Profile Selection Method: * Last Run ▾

Target Key Figures

Target Key Figure for Forecast: Most Extensive

Most Distinct Clusters

Utilize Multiple Forecasts

Method: * Choose Best Forecast ▾

Curve-based forecasting: authorization

Access to Curve clustering profiles is controlled with the Management of Planning Profiles restriction type

The **execution of Curve clustering jobs** is controlled with the Planning operators restriction type

A **new profile type** named **CURVE_BASED_FCST** should be used to refer to curve clustering profiles

Permission filters are considered via the READ access during the curve clustering process

When setting up authorization for curve clustering, please keep in mind that clustering in general **works best with larger amount of input data**

Management of Planning Profiles

PLAREA_PLPROF_PRNAME

Values	Description	Business Catalogs
Management of Planning Profiles		
Restrictions and Values		
☒ Write, Read ☐ Read		
Planning Area		ADAM
Planning Profile		ADAMMONTHAUTO1
Planning Profile Type		CURVE_BASED_FCST
☐ Write, Read ☒ Read		
Planning Area		ADAM
Planning Profile		ADAMMONTHAUTO1
Planning Profile Type		CURVE_BASED_FCST

Planning Operators

PLAREA_PLOPTY_PLOPID

Values	Description	Business Catalogs
Planning Operators		
Restrictions and Values		
☒ Write		
Planning Area		ADAM
Planning Operator ID		ADAMMONTHAUTO1
Planning Operator Type		CURVE_BASED_FCST

Curve-based forecasting: other considerations

Tip: Different algorithms work best for different planning objects. Test the new method by adding it to a best-fit model

When is curve-based forecasting exceptionally advantageous?

- When calculating forecast for longer term based on little historical data
- When a significant change in the life cycle is due in the forecast horizon
 - For example, when the history shows a monotonous upward trend, but the product is expected to start phasing out in the near future

When is curve-based forecasting less powerful

- In case of sporadic demand. Sporadic patterns are often atypical, making clustering less efficient
- When the beginning or end of the reference curves or the own history is not available
 - Using the Curve-base method combined with Product Lifecycle Management is not recommended

Restrictions

- For performance considerations, the maximum curve length in clustering is 19 years, or 39 quarters, or 59 months or 155 weeks. Daily granularity is not supported
- The reference curves must be long enough to cover the own history plus the forecast horizon, otherwise the last available forecast value will be copied to the uncovered time periods
- Planning versions and scenarios are not supported

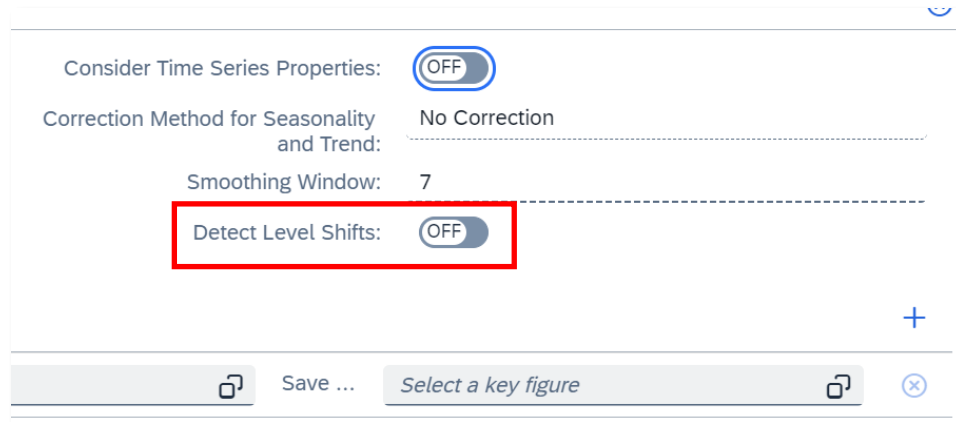
A blue geometric shape, resembling a parallelogram or a trapezoid, is positioned on the left side of the slide. It has a diagonal cut on its right side.

**Demand
Enhancements Outlier Correction**

Enhanced Outlier Correction

Level shift detection

Available for correction with smoothing, mean, median and tolerance
If “Consider Time Series Properties” is ON and Time Series Results are available, then level shift detection is skipped.



Consider Time Series Properties: ☐ OFF

Correction Method for Seasonality and Trend: No Correction

Smoothing Window: 7

Detect Level Shifts: ☐ OFF

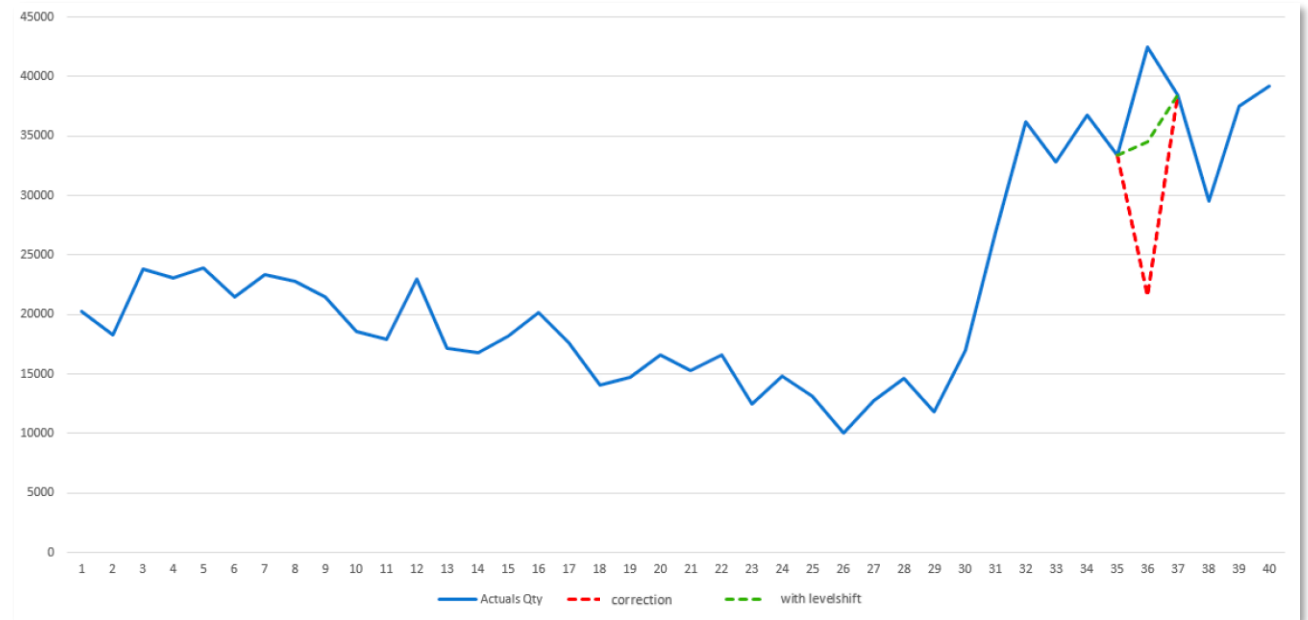
Save ... Select a key figure

With Level Shift Detection

The time series is split into segments with homogenous subsets.

Without Level Shift Detection

Substituting outliers with the mean or median of the entire dataset may introduce additional noise, or distort the seasonal pattern and the slope



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Demand

New algorithm grouping in UI

Add Forecasting Algorithm

Search

ARIMA

Auto-ARIMA/SARIMA

Auto-ARIMAX/SARIMAX

Average

Simple Average

Simple Moving Average

Weighted Average

Weighted Moving Average

Croston

Croston Method

Croston TSB Method

Demand Sensing

Demand Sensing with Gradient Boosting

Demand Sensing with Multiple Linear Regression (Full)

Demand Sensing with Multiple Linear Regression (Update)

Exponential Smoothing

Adaptive-Response-Rate Single Exponential Smoothing

Automated Exponential Smoothing

Brown Exponential Smoothing

Cancel

Add Forecasting Algorithm

Search

Demand Sensing with Multiple Linear Regression (Full)

Demand Sensing with Multiple Linear Regression (Update)

Exponential Smoothing

Adaptive-Response-Rate Single Exponential Smoothing

Automated Exponential Smoothing

Brown Exponential Smoothing

Double Exponential Smoothing

Single Exponential Smoothing

Triple Exponential Smoothing

Gradient Boosting

Gradient Boosting of Decision Trees

Linear Regression

Multiple Linear Regression

Seasonal Linear Regression

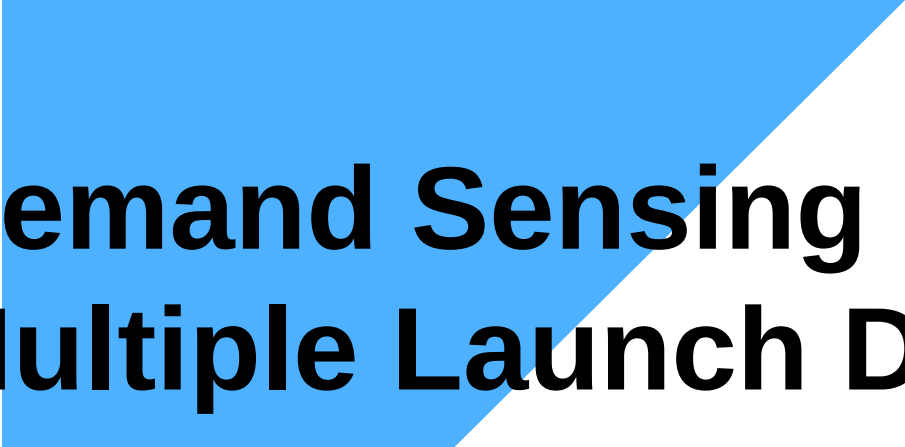
Other Algorithms

Copy Past Periods

Curve-Based Method

Daily Disaggregation Optimization

Cancel

A blue geometric shape, resembling a parallelogram with a diagonal cut, serves as a background for the text.

Demand Sensing Multiple Launch Dimensions

Multiple Launch Dimensions – Also working with Demand Sensing

Before

You could define phase-in and phase-out dates on one additional level (e.g. Location region)

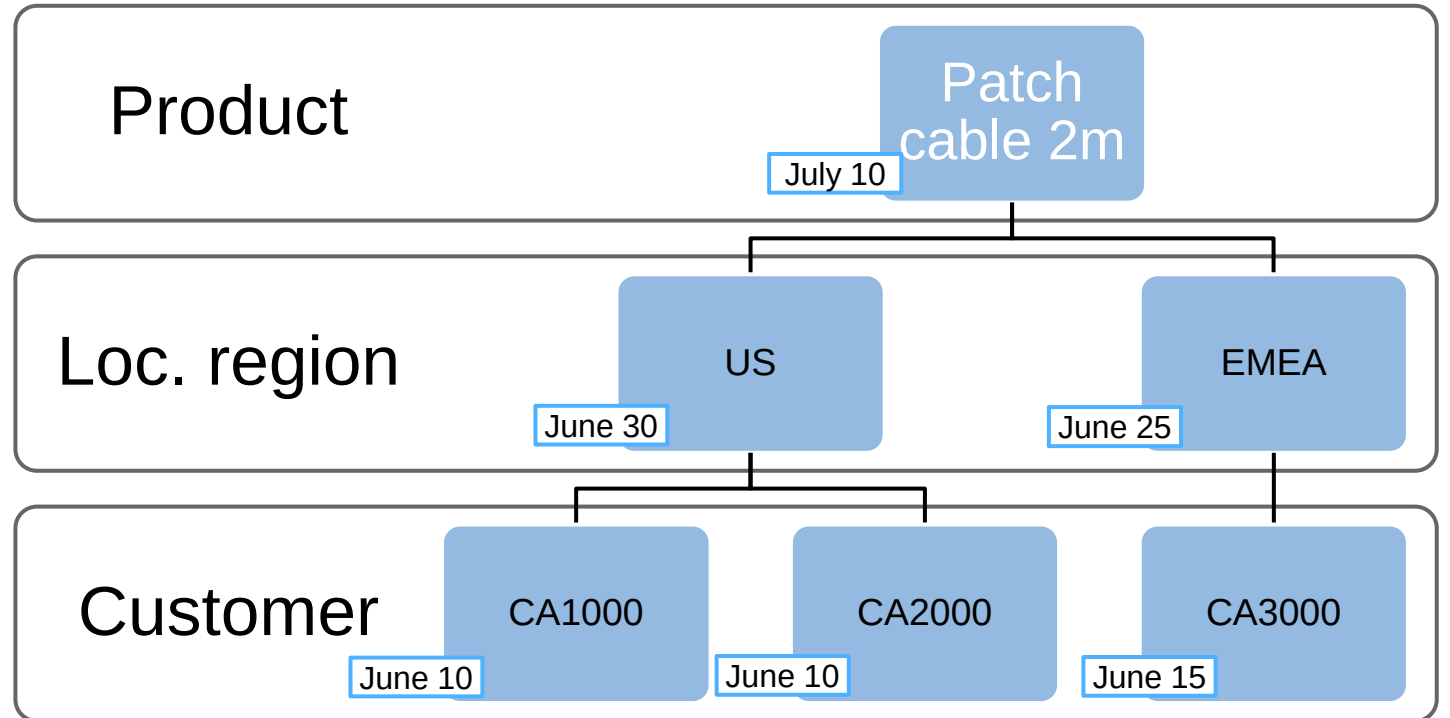
Now

You can define phase-in and phase-out dates on two additional levels (e.g. Location region and Customer)

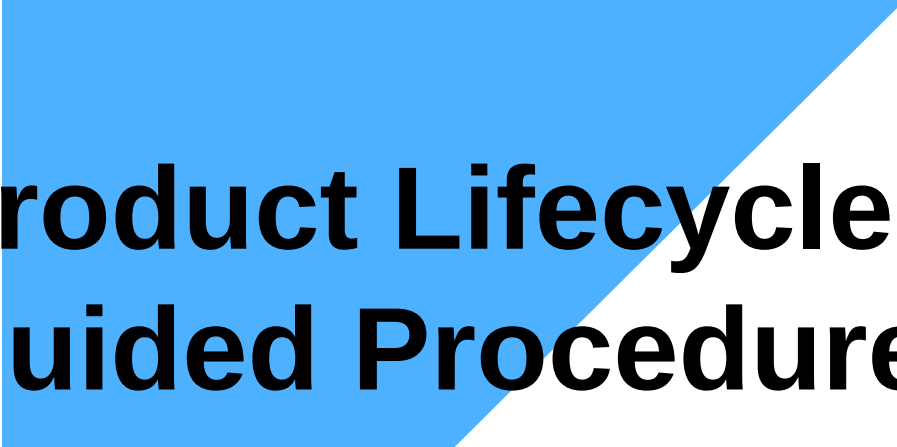
Aggregated Phase-in and Phase-out planning as well

Works with Manual Forecasting except for the parameter check

Enabled / disabled on Planning Area level



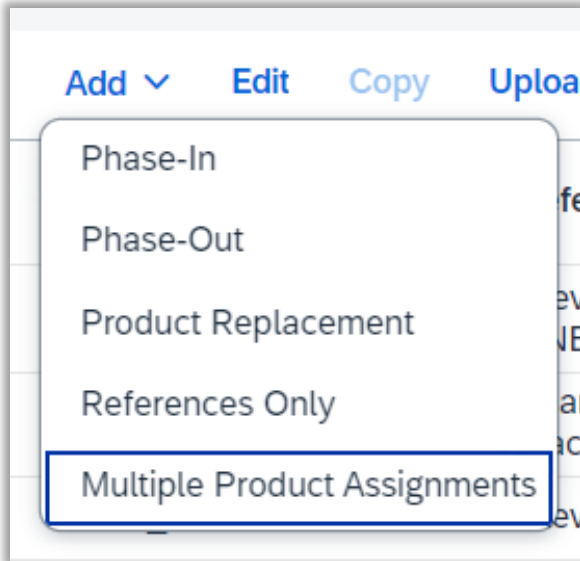
Example: new product introduction dates for regions and customers

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Product Lifecycle Management Guided Procedure

New Guided Procedure UI

A new guided UI to maintain
Single product reference assignments
and phase-in/out dates



SAP Manage Product Lifecycle

Product Replacement

Progress: 1. Product Selection — 2. Reference Assignments — 3. Reference Details — 4. Launch Dimension — 5. Phase-In Details New Product — 6. Phase-Out Details Old Product

1. Product Selection

Planning Area: WDFSAP6NP

New Product ID:

Old Product ID:

[Next Step](#) [Cancel](#)

Demo

A blue geometric graphic consisting of a rectangle with a diagonal cut from the top-left corner to the bottom-right corner, creating a trapezoidal shape on the left side of the text.

Product Lifecycle Management

Deprecation of external Master Data

Deactivation of external Master Data types for PLM

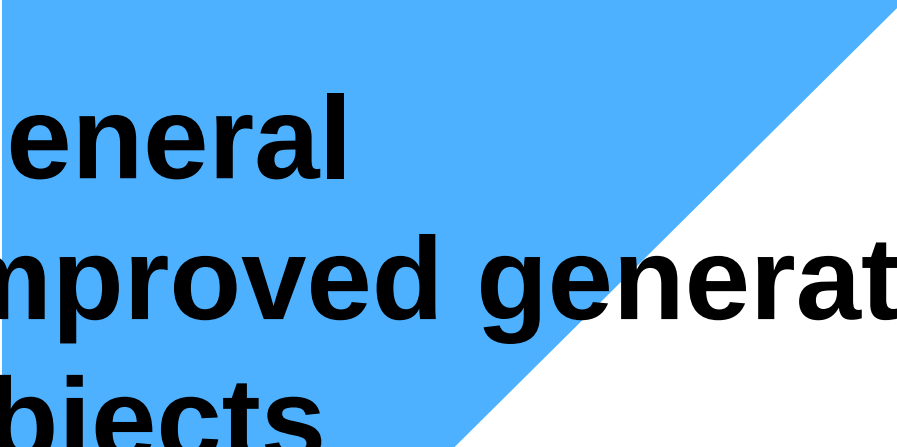
Affected Master Data Types

SMD_FCSTASSIGN, SMD_FCSTASSIGN_V2,
SMD_FCSTDATES, SMD_FCSTDATES_V2,
SMD_FCSTASSIGN_DATES

- With 2311: we will prevent the activation of external master data types with product lifecycle management data sources in systems where there were no existing external master data types. A warning is given when activating the planning area if external MDTs are assigned.
- With 2402: customers who did not use external MDTs in production system before 2402 will not be able to activate and use it in test system either
- With 2408: no activation and use of external MDTs. Planning area with external MDTs can no longer be activated.

As alternative function, affected customers either can use the

- Product Lifecycle Management App "Manage Product Lifecycle", and view the Product Assignments and Forecast Dates per navigation from the IBP Excel Add-In.
- or the OData Service from communication scenario [SAP_COM_0072](#) to see or retrieve the corresponding information.

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**General
Improved generation of missing planning
objects**

Improved Generation of missing Planning Object Records

New

global configuration parameter
ADD_ALL_PLAN_OBJECT_RECORDS to
control how missing planning object
records are created during disaggregation
in the IBP Excel Add-In or Planner
Workspaces.

Value	Effect
NO (default)	Missing planning object records are only generated if no planning object records exist that can be used as disaggregation target
YES	Missing planning object records are always generated if needed

Improved Generation of missing Planning Object Records - Example

A key figure is stored on monthly level. Its values are equally disaggregated without proportionality.

	January	February	March
Product A	<i>no record exists</i>	100	<i>no record exists</i>

Starting Point

Value of first quarter is changed from 100 to 150

	January	February	March
Product A	<i>no record exists</i>	150	<i>no record exists</i>

ADD_ALL_PLAN_OBJECT_RECORDS = 'NO'

	January	Februar	March
Product A	50	50	50

ADD_ALL_PLAN_OBJECT_RECORDS = 'YES'

Thank you.

Contact information:

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